

kirsch

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1 Tunnel Convergence Problem

1.1 Solving Kirsch's Problem Using the Release Nodal Force Approach

1.1.1 1. Introduction

Tunnel excavation redistributes the in-situ stress and deforms the surrounding rock. The inward movement of the excavation boundary is commonly called tunnel convergence. Its magnitude depends on the initial stress, tunnel geometry, constitutive behaviour, and any support traction acting on the wall.

This notebook models a circular opening in an infinite, homogeneous, isotropic, linear-elastic medium under plane-strain conditions. Excavation is represented by gradually withdrawing the traction that the removed material exerted on the remaining rock. The final in-plane stresses are verified against the analytical Kirsch solution [1], while the displacement history illustrates excavation-induced convergence.

This notebook models a two-dimensional quarter section of a circular tunnel with a radius of 6.5 m. A vertical compressive far-field stress of 20 MPa is applied, while symmetry boundary conditions are imposed along the left and bottom boundaries. The right boundary is traction-free, consistent with the uniaxial vertical far-field stress assumed for this verification benchmark.

Two modelling approaches are compared:

1. **Release nodal force approach:** the rock mass is initialized with an in-situ stress field, and the equivalent nodal forces acting at the tunnel wall are gradually released to represent excavation.
2. **Standard approach:** the tunnel is present from the beginning, and the external load is applied directly to an initially unstressed model.

Both approaches are evaluated using linear and quadratic finite elements. Stress profiles are checked along the horizontal ($\theta = 0^\circ$) and vertical ($\theta = 90^\circ$) symmetry axes. OGS [2] uses the tension-positive sign convention, so the applied compressive stress is $\sigma_\infty = -20$ MPa. The finite numerical domain extends about 10.8 tunnel radii from the opening; small far-field differences from the infinite-domain solution are therefore expected.

Learning objectives

- configure a plane-strain `SMALL_DEFORMATION` process in OGS;
- understand compensation of the non-equilibrium initial residual;
- verify numerical stresses against the Kirsch solution; and
- distinguish element-order error, domain-truncation error, and nodal stress extrapolation effects.

Release nodal force boundary condition This boundary condition applies a time-dependent *release nodal force* to the nodes of the boundary mesh on which it is defined, here the tunnel wall (`arc`). It is the numerical counterpart of the excavation process: instead of removing elements, the forces that the excavated material exerted on the remaining rock mass are gradually withdrawn.

Finite-element formulation The small-deformation process solves the discrete equilibrium equations

$$r(u, t) = F^{\text{int}}(u) - F^{\text{ext}}(t) = 0, \quad F^{\text{int}}(u) = \int_{\Omega} \mathbf{B}^T \sigma \, d\Omega,$$

where $\mathbf{B}$ is the strain–displacement matrix and F^{ext} collects the applied loads (here the far-field traction on the top boundary). With the process initialized by the in-situ stress, the total stress is

$$\sigma = \sigma_0 + \mathbb{C} : \varepsilon(u),$$

with σ_0 the in-situ stress and $\mathbb{C}$ the elasticity tensor. For plane strain, $\varepsilon_{zz} = 0$. Therefore the compatible out-of-plane initial stress is $\sigma_{0,zz} = \nu(\sigma_{0,xx} + \sigma_{0,yy}) = -6$ MPa.

Step 1 — non-equilibrium initial residual. At $t = 0$, with $u_0 = 0$, the domain containing the (traction-free) opening is *not* in equilibrium under σ_0 . OGS assembles this initial out-of-balance force once:

$$r^{\text{neq}} = r(u_0, 0) = \int_{\Omega} \mathbf{B}^T \sigma_0 \, d\Omega - F^{\text{ext}}(0).$$

The vector r^{neq} is called the **non-equilibrium initial residual**: the out-of-balance nodal force vector obtained by evaluating the equilibrium equations for the *undeformed* initial state. It is non-zero because a stress field that was in equilibrium in the intact ground is no longer in equilibrium in a domain containing the opening — at the traction-free tunnel wall nothing pushes back against σ_0 . Assembling $\int_{\Omega} \mathbf{B}^T \sigma_0 \, d\Omega$ node by node:

- at **interior nodes**, the contributions of neighbouring elements cancel — the entries vanish;
- at the **outer boundaries**, the internal forces are balanced by the applied top traction and by reactions at constrained symmetry degrees of freedom;

- at the **tunnel wall nodes**, nothing cancels them — the remaining entries are exactly the *equivalent nodal forces* that the excavated core exerted on the tunnel wall,

$$F_{0,i} = r_i^{\text{neq}} = \int_{\Gamma_{\text{arc}}} N_i (\sigma_0 \cdot n) d\Gamma, \quad i \in \text{nodes of } \Gamma_{\text{arc}},$$

with N_i the nodal shape function and n the outward normal of the wall. The `ReleaseNodalForce` boundary condition stores these values $F_{0,i}$ for every node of its boundary mesh. This step requires `<compensate_non_equilibrium_initial_residuum>true</compensate_non_equilibrium_initial_residuum>` on the process variable, since r^{neq} is only computed (and kept) for compensated variables.

Step 2 — compensation (in-situ equilibrium). In every time step the solver subtracts the stored initial residual from the assembled one,

$$\tilde{r}(u, t) = r(u, t) - r^{\text{neq}},$$

so that at $t = 0$ the model is exactly in equilibrium with σ_0 and $u = 0$: the compensation acts like a support force F_0 that still holds the tunnel wall in place — the not-yet-excavated state.

Step 3 — time-dependent release. The boundary condition then adds, as a natural (Neumann-type) contribution at each wall degree of freedom i , the load

$$b_i += -(1 - f(t)) F_{0,i},$$

which, combined with Step 2, gives the effective residual at the wall nodes

$$\tilde{r}_i(u, t) = F_i^{\text{int}}(u) - F_i^{\text{ext}}(t) - f(t) F_{0,i} = 0.$$

The wall is therefore supported by the fictitious force $f(t) F_0$, controlled by the user-defined *time-decay function* $f(t)$:

- at $t = 0$: $f = 1$, the full in-situ support acts, $u = 0$ (no deformation);
- for $0 < t < t_e$: partial release of the support-traction field $t_{\text{support}}(\theta, t) = f(t) \sigma_0 n(\theta)$;
- at $t \geq t_e$: $f = 0$, the wall is completely unloaded (traction-free) — the fully excavated, unsupported tunnel of the Kirsch problem.

Decay function used here A linear decay over the excavation duration $t_e = 2 \text{ d} = 1.728 \times 10^5 \text{ s}$ is used:

$$f(t) = \begin{cases} 1 - \frac{t}{t_e}, & 0 \leq t \leq t_e, \\ 0, & t > t_e. \end{cases}$$

Project-file definition In the project file, the boundary condition is defined on the process variable displacement as:

```
<boundary_condition>
  <mesh>arc</mesh>
  <type>ReleaseNodalForce</type>
  <time_decay_parameter>decay_function</time_decay_parameter>
</boundary_condition>
```

where `decay_function` is a Function-type parameter implementing $f(t)$ above. Under the uniaxial initial stress used here this is a spatially varying vector traction, not a uniform scalar pressure. The method requires `<initial_stress>` and `compensate_non_equilibrium_initial_residuum`.

```
[1]: import os
from pathlib import Path
import shutil
import sys

import gmsh
import matplotlib.pyplot as plt
import numpy as np
import ogstools as ot
import pyvista as pv
from mesh_tunnel import MeshGenerator
from prj_tunnel import PrjGenerator
```

1.1.2 2. Model parameters

Physical and numerical parameters are defined here and passed explicitly to the mesh and project generators. The absolute value of `TUNNEL_CENTER_Y` only locates the mesh; gravity is disabled and the in-situ stress is prescribed.

```
[2]: # Kirsch benchmark parameters.
YOUNGS_MODULUS = 5.0e9 # Pa
POISSONS_RATIO = 0.30
VERTICAL_STRESS = 20.0e6 # Pa, positive magnitude
TUNNEL_RADIUS = 6.5 # m
TUNNEL_CENTER_X = 0.0 # m
TUNNEL_CENTER_Y = -857.0 # m
DOMAIN_EXTENT = 70.0 # m from each symmetry axis
INNER_BLOCK_EXTENT = 35.0 # m
MESH_TRANSITION_WIDTH = 5.0 # m outside the tunnel wall
EXCAVATION_DURATION = 1.728e5 # s (2 days)
END_TIME = 3.456e5 # s (4 days)
```

1.1.3 3. Solver settings

Nonlinear and linear solver configuration passed to `PrjGenerator`. Change these to experiment with different solvers without touching the model parameters above.

```

[3]: # Nonlinear solver (SMALL_DEFORMATION supports only Newton).
NONLINEAR_MAX_ITER = 10

# Linear solver: pick one preset by name.
LINEAR_SOLVER = "SparseLU"

LINEAR_SOLVER_PRESETS = {
    "BiCGSTAB + ILUT": {
        "linear_solver_type": "BiCGSTAB",
        "preconditioner_type": "ILUT",
        "linear_max_iter": 10000,
        "linear_error_tolerance": 1e-12,
        "linear_solver_scaling": False,
    },
    "BiCGSTAB + DIAGONAL": {
        "linear_solver_type": "BiCGSTAB",
        "preconditioner_type": "DIAGONAL",
        "linear_max_iter": 10000,
        "linear_error_tolerance": 1e-12,
        "linear_solver_scaling": False,
    },
    "CG + DIAGONAL": {
        "linear_solver_type": "CG",
        "preconditioner_type": "DIAGONAL",
        "linear_max_iter": 10000,
        "linear_error_tolerance": 1e-12,
        "linear_solver_scaling": False,
    },
    # Direct solver: no preconditioner/iterations; optional diagonal
    # scaling (needs an OGS build with Eigen unsupported modules).
    "SparseLU + scaling": {
        "linear_solver_type": "SparseLU",
        "preconditioner_type": None,
        "linear_max_iter": None,
        "linear_error_tolerance": None,
        "linear_solver_scaling": True,
    },
    "SparseLU": {
        "linear_solver_type": "SparseLU",
        "preconditioner_type": None,
        "linear_max_iter": None,
        "linear_error_tolerance": None,
        "linear_solver_scaling": False,
    },
}

linear_solver_options = LINEAR_SOLVER_PRESETS[LINEAR_SOLVER]

```

```
print(f"Linear solver: {LINEAR_SOLVER} -> {linear_solver_options}")
```

```
Linear solver: SparseLU -> {'linear_solver_type': 'SparseLU',
'preconditioner_type': None, 'linear_max_iter': None, 'linear_error_tolerance':
None, 'linear_solver_scaling': False}
```

```
[4]: # Set this to True only when you intentionally want to erase all generated
      ↪ files.
      reset_generated_folders = False

      if reset_generated_folders:
          for folder in [Path("_out"), Path("_prj")]:
              if folder.is_dir():
                  shutil.rmtree(folder)
                  print(f"Deleted: {folder}")
```

```
[5]: out_dir = Path(os.environ.get("OGS_TESTRUNNER_OUT_DIR", "_out"))
      if not out_dir.exists():
          out_dir.mkdir(parents=True)
          print(f"Created: {out_dir}")
      mesh_dir_quad = Path("input_meshes_quadratic")
      mesh_dir_line = Path("input_meshes_linear")
      # create prj directory if it does not exist
      prj_dir = Path("_prj")
      if not prj_dir.exists():
          prj_dir.mkdir(parents=True)
          print(f"Created: {prj_dir}")
```

1.1.4 4. Mesh generation

The same structured topology is generated with quadratic and linear elements. Quadratic geometry represents the circular wall more accurately and also improves extrapolated nodal stresses near the opening.

4.1 Quadratic elements

```
[6]: if not gmsh.isInitialized():
      gmsh.initialize(readConfigFiles=False)
      gmsh.model.add(mesh_dir_quad.name)
      mesh_generator = MeshGenerator(
          gmsh_model=gmsh.model,
          tunnel_radius_i=TUNNEL_RADIUS,
          tunnel_radius_o=MESH_TRANSITION_WIDTH,
          tunnel_center_x=TUNNEL_CENTER_X,
          tunnel_center_y=TUNNEL_CENTER_Y,
          width=DOMAIN_EXTENT,
          height=DOMAIN_EXTENT,
          width_m=INNER_BLOCK_EXTENT,
          height_m=INNER_BLOCK_EXTENT,
```

```
)
mesh_generator.generate_meshes(out_dir=mesh_dir_quad, order=2)

gmsh.finalize()
```

```
INFO:ogstools.meshes.gmsh_converter:Detected domain dimension of 2
INFO:ogstools.meshes.gmsh_converter:Found material IDs: [1]
INFO:ogstools.meshes.gmsh_converter:Renumbered to: [0]
INFO:ogstools.meshes.gmsh_converter:domain: UnstructuredGrid (0x79b93825b8e0)
```

```
  N Cells:      2800
  N Points:     2921
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:     -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      1
```

```
Info      : Meshing 1D...
Info      : [ 0%] Meshing curve 1 (Line)
Info      : [ 10%] Meshing curve 2 (Line)
Info      : [ 10%] Meshing curve 3 (Line)
Info      : [ 20%] Meshing curve 4 (Line)
Info      : [ 20%] Meshing curve 5 (Line)
Info      : [ 30%] Meshing curve 6 (Line)
Info      : [ 30%] Meshing curve 7 (Line)
Info      : [ 40%] Meshing curve 8 (Line)
Info      : [ 40%] Meshing curve 9 (Line)
Info      : [ 50%] Meshing curve 10 (Line)
Info      : [ 50%] Meshing curve 11 (Line)
Info      : [ 60%] Meshing curve 12 (Line)
Info      : [ 60%] Meshing curve 13 (Line)
Info      : [ 60%] Meshing curve 14 (Line)
Info      : [ 70%] Meshing curve 15 (Line)
Info      : [ 70%] Meshing curve 16 (Line)
Info      : [ 80%] Meshing curve 17 (Circle)
Info      : [ 80%] Meshing curve 18 (Circle)
Info      : [ 90%] Meshing curve 19 (Circle)
Info      : [ 90%] Meshing curve 20 (Circle)
Info      : [100%] Meshing curve 21 (Line)
Info      : [100%] Meshing curve 22 (Line)
Info      : Done meshing 1D (Wall 0.000514383s, CPU 0.000854s)
Info      : Meshing 2D...
Info      : [ 0%] Meshing surface 1 (Transfinite)
Info      : [ 20%] Meshing surface 2 (Transfinite)
Info      : [ 30%] Meshing surface 3 (Transfinite)
Info      : [ 40%] Meshing surface 4 (Transfinite)
Info      : [ 60%] Meshing surface 5 (Transfinite)
Info      : [ 70%] Meshing surface 6 (Transfinite)
Info      : [ 80%] Meshing surface 7 (Transfinite)
```

```

Info      : [ 90%] Meshing surface 8 (Plane, Delaunay)
Info      : Done meshing 2D (Wall 0.00239339s, CPU 0.002504s)
Info      : 3082 nodes 3577 elements
Info      : Writing 'input_meshes_quadratic/domain.msh'...
Info      : Done writing 'input_meshes_quadratic/domain.msh'

info: Reordering nodes...
info: Method: Reversing order of nodes unless it is considered correct by the
OGS6 standard, i.e. such that  $\det(J) > 0$ , where J is the Jacobian of the global-
to-local coordinate transformation.
info: Corrected 0 elements.
info: VTU file written.

INFO:ogstools.meshes.gmsh_converter:bottom: UnstructuredGrid (0x79b93825be80)
  N Cells:      60
  N Points:     61
  X Bounds:     6.500e+00, 7.000e+01
  Y Bounds:    -8.570e+02, -8.570e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2
INFO:ogstools.meshes.gmsh_converter:left: UnstructuredGrid (0x79b8f919d780)
  N Cells:      60
  N Points:     61
  X Bounds:     0.000e+00, 0.000e+00
  Y Bounds:    -8.505e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2
INFO:ogstools.meshes.gmsh_converter:right: UnstructuredGrid (0x79b8f919dd20)
  N Cells:      40
  N Points:     41
  X Bounds:     7.000e+01, 7.000e+01
  Y Bounds:    -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2
INFO:ogstools.meshes.gmsh_converter:top: UnstructuredGrid (0x79b8f919df00)
  N Cells:      40
  N Points:     41
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:    -7.870e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2
INFO:ogstools.meshes.gmsh_converter:arc: UnstructuredGrid (0x79b8f919e0e0)
  N Cells:      40
  N Points:     41
  X Bounds:     0.000e+00, 6.500e+00
  Y Bounds:    -8.570e+02, -8.505e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2

```

```

INFO:ogstools.meshes.gmsh_converter:Conversion complete.

domain: 2800 cells
bottom: 60 cells
left: 60 cells
right: 40 cells
top: 40 cells
arc: 40 cells
info: Reordering nodes...
info: Method: Reversing order of nodes unless it is considered correct by the
OGS6 standard, i.e. such that  $\det(J) > 0$ , where  $J$  is the Jacobian of the global-
to-local coordinate transformation.
info: Corrected 0 elements.
info: VTU file written.
Create quadratic mesh for input_meshes_quadratic/domain.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for input_meshes_quadratic/top.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for input_meshes_quadratic/left.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for input_meshes_quadratic/arc.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for input_meshes_quadratic/right.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for input_meshes_quadratic/bottom.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
info: Mesh reading time: 0.00996176 s
info: MeshNodeSearcher construction time: 0.000297193 s
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 0.000692647 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00240334 s
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 5.3762e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00888494 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'bottom' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 5.0267e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00581341 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'left' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 2.5403e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00586768 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'right' and it is equal to the newly computed values.

```

```

info: identifySubdomainMesh(): identifySubdomainMeshNodes took 2.651e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00549851 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'top' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 0.000174025 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.0056935 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'arc' and it is equal to the newly computed values.
info: identifySubdomains time: 0.0360652 s
info: writing time: 0.0114709 s
info: Entire run time: 0.0578631 s

```

4.2 Linear elements

```

[7]: if not gmsh.isInitialized():
    gmsh.initialize(readConfigFiles=False)
gmsh.model.add(mesh_dir_line.name)
mesh_generator = MeshGenerator(
    gmsh_model=gmsh.model,
    tunnel_radius_i=TUNNEL_RADIUS,
    tunnel_radius_o=MESH_TRANSITION_WIDTH,
    tunnel_center_x=TUNNEL_CENTER_X,
    tunnel_center_y=TUNNEL_CENTER_Y,
    width=DOMAIN_EXTENT,
    height=DOMAIN_EXTENT,
    width_m=INNER_BLOCK_EXTENT,
    height_m=INNER_BLOCK_EXTENT,
)
mesh_generator.generate_meshes(out_dir=mesh_dir_line, order=1)

gmsh.finalize()

```

```

INFO:ogstools.meshes.gmsh_converter:Detected domain dimension of 2
INFO:ogstools.meshes.gmsh_converter:Found material IDs: [1]
INFO:ogstools.meshes.gmsh_converter:Renumbered to: [0]
INFO:ogstools.meshes.gmsh_converter:domain: UnstructuredGrid (0x79b8f919de40)
  N Cells:      2800
  N Points:     2921
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:     -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     1
INFO:ogstools.meshes.gmsh_converter:bottom: UnstructuredGrid (0x79b8f919e680)
  N Cells:      60
  N Points:     61
  X Bounds:     6.500e+00, 7.000e+01
  Y Bounds:     -8.570e+02, -8.570e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:     2

```

```

INFO:ogstools.meshes.gmsh_converter:left: UnstructuredGrid (0x79b8f919dc00)
  N Cells:    60
  N Points:   61
  X Bounds:   0.000e+00, 0.000e+00
  Y Bounds:   -8.505e+02, -7.870e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   2
INFO:ogstools.meshes.gmsh_converter:right: UnstructuredGrid (0x79b8f919d960)
  N Cells:    40
  N Points:   41
  X Bounds:   7.000e+01, 7.000e+01
  Y Bounds:   -8.570e+02, -7.870e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   2
INFO:ogstools.meshes.gmsh_converter:top: UnstructuredGrid (0x79b8f919e860)
  N Cells:    40
  N Points:   41
  X Bounds:   0.000e+00, 7.000e+01
  Y Bounds:   -7.870e+02, -7.870e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   2
INFO:ogstools.meshes.gmsh_converter:arc: UnstructuredGrid (0x79b8f919e980)
  N Cells:    40
  N Points:   41
  X Bounds:   0.000e+00, 6.500e+00
  Y Bounds:   -8.570e+02, -8.505e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   2
INFO:ogstools.meshes.gmsh_converter:Conversion complete.

Info      : Meshing 1D...
Info      : [ 0%] Meshing curve 1 (Line)
Info      : [ 10%] Meshing curve 2 (Line)
Info      : [ 10%] Meshing curve 3 (Line)
Info      : [ 20%] Meshing curve 4 (Line)
Info      : [ 20%] Meshing curve 5 (Line)
Info      : [ 30%] Meshing curve 6 (Line)
Info      : [ 30%] Meshing curve 7 (Line)
Info      : [ 40%] Meshing curve 8 (Line)
Info      : [ 40%] Meshing curve 9 (Line)
Info      : [ 50%] Meshing curve 10 (Line)
Info      : [ 50%] Meshing curve 11 (Line)
Info      : [ 60%] Meshing curve 12 (Line)
Info      : [ 60%] Meshing curve 13 (Line)
Info      : [ 60%] Meshing curve 14 (Line)
Info      : [ 70%] Meshing curve 15 (Line)
Info      : [ 70%] Meshing curve 16 (Line)
Info      : [ 80%] Meshing curve 17 (Circle)

```

```

Info      : [ 80%] Meshing curve 18 (Circle)
Info      : [ 90%] Meshing curve 19 (Circle)
Info      : [ 90%] Meshing curve 20 (Circle)
Info      : [100%] Meshing curve 21 (Line)
Info      : [100%] Meshing curve 22 (Line)
Info      : Done meshing 1D (Wall 0.000461795s, CPU 0.000745s)
Info      : Meshing 2D...
Info      : [  0%] Meshing surface 1 (Transfinite)
Info      : [ 20%] Meshing surface 2 (Transfinite)
Info      : [ 30%] Meshing surface 3 (Transfinite)
Info      : [ 40%] Meshing surface 4 (Transfinite)
Info      : [ 60%] Meshing surface 5 (Transfinite)
Info      : [ 70%] Meshing surface 6 (Transfinite)
Info      : [ 80%] Meshing surface 7 (Transfinite)
Info      : [ 90%] Meshing surface 8 (Plane, Delaunay)
Info      : Done meshing 2D (Wall 0.00221872s, CPU 0.001581s)
Info      : 3082 nodes 3577 elements
Info      : Writing 'input_meshes_linear/domain.msh'...
Info      : Done writing 'input_meshes_linear/domain.msh'

```

info: Reordering nodes...

info: Method: Reversing order of nodes unless it is considered correct by the OGS6 standard, i.e. such that $\det(J) > 0$, where J is the Jacobian of the global-to-local coordinate transformation.

info: Corrected 0 elements.

info: VTU file written.

domain: 2800 cells

bottom: 60 cells

left: 60 cells

right: 40 cells

top: 40 cells

arc: 40 cells

info: Reordering nodes...

info: Method: Reversing order of nodes unless it is considered correct by the OGS6 standard, i.e. such that $\det(J) > 0$, where J is the Jacobian of the global-to-local coordinate transformation.

info: Corrected 0 elements.

info: VTU file written.

info: Mesh reading time: 0.00590838 s

info: MeshNodeSearcher construction time: 0.000152698 s

info: identifySubdomainMesh(): identifySubdomainMeshNodes took 0.000269807 s

info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00169759 s

info: identifySubdomainMesh(): identifySubdomainMeshNodes took 8.561e-06 s

info: There is already a 'bulk_node_ids' property present in the subdomain mesh 'bottom' and it is equal to the newly computed values.

info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.000455619 s

info: There is already a 'bulk_element_ids' property present in the subdomain mesh 'bottom' and it is equal to the newly computed values.

```

info: identifySubdomainMesh(): identifySubdomainMeshNodes took 2.3752e-05 s
info: There is already a 'bulk_node_ids' property present in the subdomain mesh
'left' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.000462049 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'left' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.3452e-05 s
info: There is already a 'bulk_node_ids' property present in the subdomain mesh
'right' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00044761 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'right' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.1532e-05 s
info: There is already a 'bulk_node_ids' property present in the subdomain mesh
'top' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.000666194 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'top' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.7806e-05 s
info: There is already a 'bulk_node_ids' property present in the subdomain mesh
'arc' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.000462127 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'arc' and it is equal to the newly computed values.
info: identifySubdomains time: 0.00493665 s
info: writing time: 0.00787543 s
info: Entire run time: 0.0190024 s

```

```

[8]: kirsch_project_file = Path("kirsch.prj")
prj_generator = PrjGenerator(
    mesh_dir=Path(), # mesh files are selected through ot.Model / OGS -m
    element_order=2,
    youngs_modulus=YOUNGS_MODULUS,
    poissons_ratio=POISSONS_RATIO,
    vertical_stress=VERTICAL_STRESS,
    tunnel_radius=TUNNEL_RADIUS,
    tunnel_center_y=TUNNEL_CENTER_Y,
    excavation_duration=EXCAVATION_DURATION,
    end_time=END_TIME,
    nonlinear_max_iter=NONLINEAR_MAX_ITER,
    **linear_solver_options,
)
prj = prj_generator.generate_project(output_file=kirsch_project_file)

```

```

[9]: m = ot.Model(
    project=prj,
    meshes=mesh_dir_quad,

```

```

)

colors = {
    "arc": "#26E600",
    "bottom": "#56B4E9",
    "left": "#009E73",
    "right": "#CC79A7", # appears only if it has an explicit BC
    "top": "#D55E00",
}

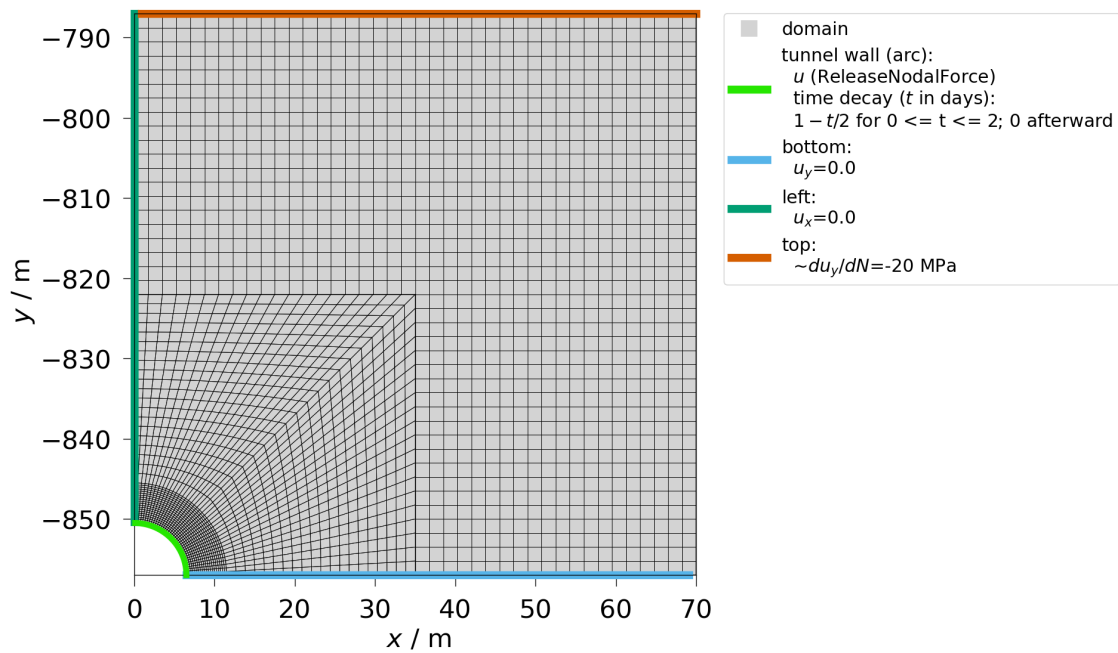
fig = m.plot_constraints(figsize=(16, 9), fontsize=26, leg_fontsize=18)
ax = fig.axes[0]

# Change boundary colors and emphasize the tunnel wall.
for line in ax.lines:
    name = line.get_label().split(":", 1)[0]
    if name in colors:
        line.set_color(colors[name])
    if name == "arc":
        line.set_linewidth(6)
        line.set_zorder(10)

# Display the project values in engineering units in this figure.
excavation_duration_days = EXCAVATION_DURATION / (24 * 60 * 60)
vertical_stress_mpa = VERTICAL_STRESS / 1.0e6

# Change matching legend colors and replace the raw SI values.
legend = ax.get_legend()
for handle, text in zip(legend.legend_handles, legend.get_texts()):
    name = text.get_text().split(":", 1)[0]
    if name in colors:
        handle.set_color(colors[name])
    label = text.get_text()
    if name == "arc":
        label = (
            "tunnel wall (arc):\n"
            "  $u$ (ReleaseNodalForce)\n"
            "  time decay ($t$ in days):\n"
            f"  $1-t/{excavation_duration_days:g}$ for "
            f"$0 \le t \le {excavation_duration_days:g}$; "
            "0 afterward"
        )
    elif name == "top":
        label = label.replace(
            f"-{VERTICAL_STRESS}", f"-{vertical_stress_mpa:g} MPa"
        )
    text.set_text(label)

```



1.1.5 5. Computation

```
[10]: def runBenchmark(
    project_file,
    out_dir,
    prj_dir,
    output_prefix,
    mesh_dir,
    order=2,
    standard_approach=False,
    use_element_deactivation=False,
    ogs_path=None,
):
    prj = ot.Project(
        input_file=project_file,
        output_file=Path(prj_dir, f"{output_prefix}.prj"),
    )

    xpath_var = './process_variables/process_variable[name="displacement"]/'
    prj.replace_text(
        order,
        xpath=xpath_var + "order",
    )

    prj.replace_text(
        output_prefix,
```

```

        xpath="./time_loop/output/prefix",
    )

    if standard_approach:
        prj.replace_text(
            "false",
            xpath=xpath_var + "compensate_non_equilibrium_initial_residuum",
        )
        prj.remove_element(
            xpath=xpath_var + 'boundary_conditions/
↳boundary_condition[mesh="arc"] '
        )
        prj.remove_element(xpath="./processes/process/initial_stress")

        prj.remove_element(xpath="./time_loop/processes/process/time_stepping")
        prj.time_loop.set_stepping(
            process="SD",
            type="FixedTimeStepping",
            t_initial="0",
            t_end="1",
            repeat="1",
            delta_t="1.0",
        )

    if use_element_deactivation:
        prj.add_block(
            blocktag="deactivated_subdomains",
            parent_xpath="./process_variables/process_variable",
        )
        prj.add_block(
            blocktag="deactivated_subdomain",
            parent_xpath="./process_variables/process_variable/
↳deactivated_subdomains",
            taglist=["time_interval", "material_ids"],
            textlist=["", "1"],
        )
        prj.add_element(
            parent_xpath="./process_variables/process_variable/
↳deactivated_subdomains/deactivated_subdomain/time_interval",
            tag="start",
            text="0",
        )
        prj.add_element(
            parent_xpath="./process_variables/process_variable/
↳deactivated_subdomains/deactivated_subdomain/time_interval",
            tag="end",
            text="1e+18",
        )

```

```

    )

    prj.write_input()

    # Prefer an explicit path, then the active environment, then PATH.
    if ogs_path is not None:
        candidate = Path(ogs_path)
        ogs_executable = candidate if candidate.name == "ogs" else candidate / ␣
↪ "ogs"
    else:
        environment_candidate = Path(sys.executable).parent / "ogs"
        path_candidate = shutil.which("ogs")
        ogs_executable = (
            environment_candidate
            if environment_candidate.is_file()
            else Path(path_candidate) if path_candidate else ␣
↪ environment_candidate
        )
    if not ogs_executable.is_file():
        raise FileNotFoundError(
            "OGS executable not found. Install it in the active environment, "
            "put it on PATH, or pass ogs_path to runBenchmark()."
        )

    execution = ot.Execution(
        ogs_path=str(ogs_executable.parent),
        omp_num_threads=4,
        ogs_asm_threads=4,
        write_logs=True,
    )

    model = ot.Model(project=prj, meshes=mesh_dir, execution=execution)
    simulation = model.run(
        target=Path(out_dir, output_prefix),
        overwrite=True,
    )

    print(f"Output prefix is {output_prefix}")
    print(f"OGS executable: {ogs_executable}")

    return simulation.meshseries_file

```

5.1 Release nodal force approach

```

[11]: # This generated project contains the initial stress, arc release, and
      # time-decay configuration used by the ReleaseNodalForce simulation.
      release_project_file = kirsch_project_file
      pvd = runBenchmark(

```

```

    project_file=release_project_file,
    out_dir=out_dir,
    prj_dir=prj_dir,
    output_prefix="kirsch",
    mesh_dir=mesh_dir_quad,
)

```

Output prefix is kirsch

OGS executable: /home/mehran/.venvs/ogs312/bin/ogs

```

[12]: ms = ot.MeshSeries(pvd).scale(time="d")
      ms_last = ms[-1]
      xs = np.linspace(TUNNEL_RADIUS, DOMAIN_EXTENT, 35)
      probes = np.c_[
          TUNNEL_CENTER_X + xs,
          np.full_like(xs, TUNNEL_CENTER_Y),
          np.zeros_like(xs),
      ]
      extracted_ms = ot.MeshSeries.probe(ms, probes)

```

100%| | 22/22 [00:00<00:00, 44.99it/s]

```

[13]: pvd_linear = runBenchmark(
      project_file=release_project_file,
      out_dir=out_dir,
      prj_dir=prj_dir,
      output_prefix="kirsch_linear_e",
      mesh_dir=mesh_dir_line,
      order=1,
)

```

Output prefix is kirsch_linear_e

OGS executable: /home/mehran/.venvs/ogs312/bin/ogs

```

[14]: ms_linear = ot.MeshSeries(pvd_linear).scale(time="d")
      extracted_ms_linear = ot.MeshSeries.probe(ms_linear, probes)

```

100%| | 22/22 [00:00<00:00, 101.31it/s]

```

[15]: ot.plot.setup.num_levels = 6
      displacement = ot.variables.displacement
      fig, ax = plt.subplots(1, 3, figsize=(21, 4), gridspec_kw={"wspace": 0.5})
      for axis, variable in zip(ax, [displacement["x"], displacement["y"],
      ↪displacement]):
          ot.plot.contourf(
              ms_last,
              variable,
              fig=fig,
              ax=axis,

```

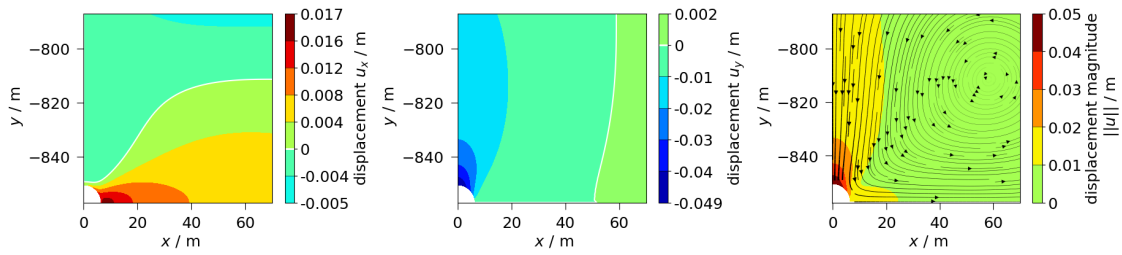
```

        fontsize=18,
        cb_labelsize=18,
        cmap="jet",
    )

    cb_length, cb_width = 1.0, 1.0 # fractions of the default colorbar size
    for cax in [a for a in fig.axes if a.get_label() == "<colorbar>"]:
        p = cax.get_position()
        h = p.height * cb_length
        cax.set_position([p.x0, p.y0 + 0.5 * (p.height - h), p.width * cb_width, h])

plt.show()

```



The final stress state is shown as Cartesian and polar components. The stress tensor is rotated into a polar coordinate system centred at the tunnel axis,

$$\sigma' = \mathbf{R}^T \sigma \mathbf{R},$$

which is done directly by `ogstools` via `ot.variables.stress.to_polar(center=...)`; the radial and tangential components are then simply accessed as `["rr"]` and `["tt"]`.

The analytical traction condition gives $\sigma_{rr} = 0$ at the wall. Postprocessed OGS values are extrapolated from integration points to nodes, so a small nonzero wall value is expected, especially for linear elements.

```

[16]: stress = ot.variables.stress
      stress_polar = stress.to_polar(center=(TUNNEL_CENTER_X, TUNNEL_CENTER_Y, 0.0))

      fig, ax = plt.subplots(
          2, 2, figsize=(12, 8), gridspec_kw={"wspace": 0.4, "hspace": 0.3}
      )
      variables = [stress["xx"], stress["yy"], stress_polar["rr"], stress_polar["tt"]]
      for axis, variable in zip(ax.ravel(), variables):
          ot.plot.contourf(
              ms_last,
              variable,
              fig=fig,

```

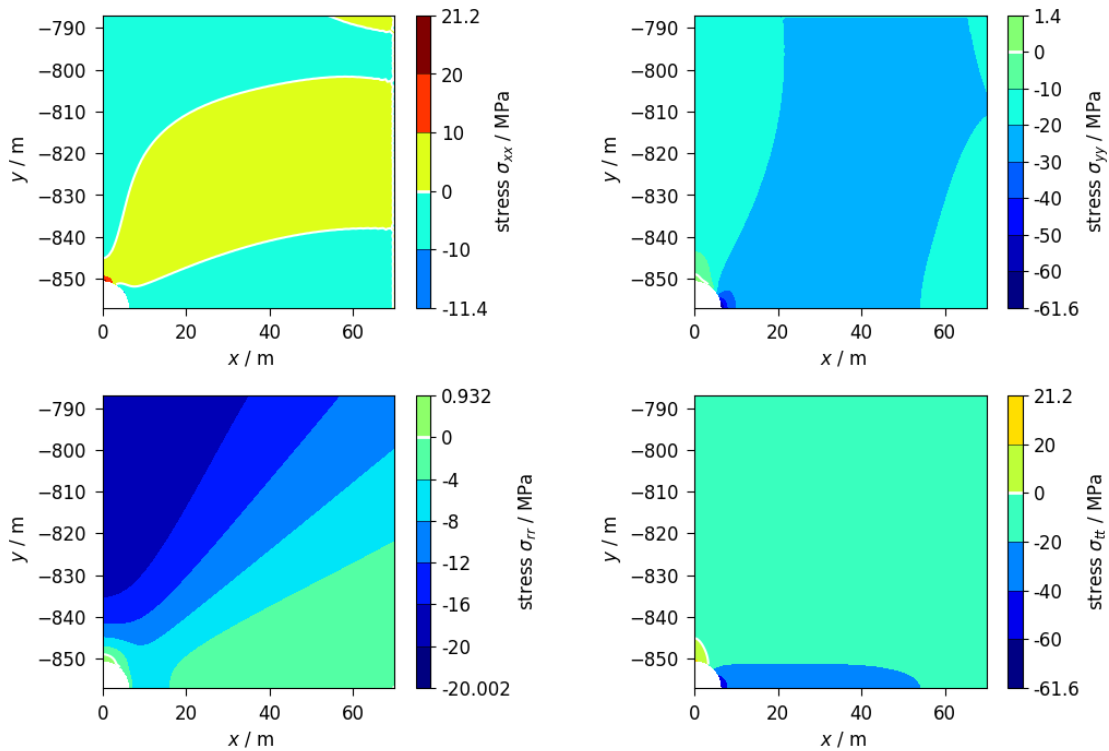
```

    ax=ax,
    fontsize=12,
    cb_labelsize=12,
    cmap="jet",
)

fig.canvas.draw() # finalize the axes positions before aligning colorbars
colorbar_axes = [
    axis for axis in fig.axes if axis.get_label() == "<colorbar>"
]
for plot_axis, colorbar_axis in zip(ax.ravel(), colorbar_axes):
    plot_pos = plot_axis.get_position()
    bar_pos = colorbar_axis.get_position()
    colorbar_axis.set_position(
        [bar_pos.x0, plot_pos.y0, bar_pos.width, plot_pos.height]
    )

plt.show()

```



For a frictional material such as rock, proximity to failure is governed by two invariants together: a deviatoric measure and the mean stress. The octahedral shear stress,

$$\tau_{\text{oct}} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} = \sqrt{\frac{2}{3}} J_2,$$

is pressure-independent and shows where distortional (shear) loading concentrates, while the mean stress,

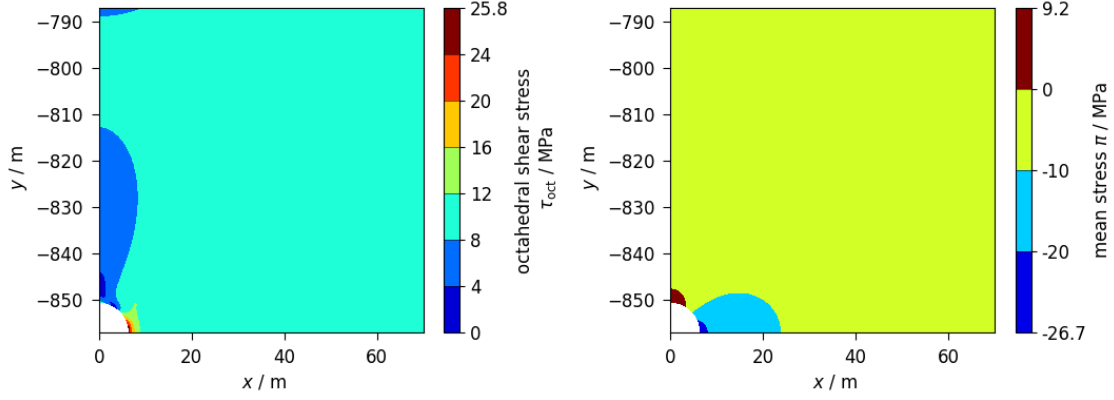
$$\bar{\sigma} = \frac{1}{3} \text{tr } \sigma,$$

describes the hydrostatic part of the three-dimensional plane-strain stress state. These invariants help interpret where excavation changes shear loading and mean stress. They are diagnostic quantities only: assessing failure with Mohr–Coulomb or Hoek–Brown would additionally require strength parameters and a consistent compression-positive or tension-positive convention.

```
[17]: fig, ax = plt.subplots(1, 2, figsize=(12, 4), gridspec_kw={"wspace": 0.4})
      for axis, variable in zip(ax, [stress.octahedral_shear, stress.tensor_mean]):
          ot.plot.contourf(
              ms_last,
              variable,
              fig=fig,
              ax=axis,
              fontsize=12,
              cb_labelsize=12,
              cmap="jet",
          )

      fig.canvas.draw() # finalize the axes positions before aligning colorbars
      colorbar_axes = [
          axis for axis in fig.axes if axis.get_label() == "<colorbar>"
      ]
      for plot_axis, colorbar_axis in zip(ax, colorbar_axes):
          plot_pos = plot_axis.get_position()
          bar_pos = colorbar_axis.get_position()
          colorbar_axis.set_position(
              [bar_pos.x0, plot_pos.y0, bar_pos.width, plot_pos.height]
          )

      plt.show()
```



In addition to the contour maps, the polar stress components are shown radially, i.e. as profiles over the distance r from the tunnel axis along the two symmetry lines $\theta = 0^\circ$ (horizontal) and $\theta = 90^\circ$ (vertical). They are compared with the full Kirsch solution [1] for a uniaxial far-field stress σ_∞ acting in the vertical direction,

$$\sigma_{rr}(r, \theta) = \frac{\sigma_\infty}{2} \left(1 - \frac{a^2}{r^2} \right) - \frac{\sigma_\infty}{2} \left(1 - \frac{4a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta,$$

$$\sigma_{\theta\theta}(r, \theta) = \frac{\sigma_\infty}{2} \left(1 + \frac{a^2}{r^2} \right) + \frac{\sigma_\infty}{2} \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta.$$

The polar components are evaluated with `stress_polar.transform` on the 2D domain mesh (where `to_polar` performs the in-plane rotation) and then sampled along the two radial lines.

The Cartesian components σ_{xx} and σ_{yy} are shown in the first row, followed by the polar components σ_{rr} and $\sigma_{\theta\theta}$ in the second row. On the two symmetry lines the shear component $\sigma_{r\theta}$ vanishes, so the Cartesian components coincide with the polar ones there: along $\theta = 0^\circ$, $\sigma_{xx} = \sigma_{rr}$ and $\sigma_{yy} = \sigma_{\theta\theta}$, while along $\theta = 90^\circ$ the roles are swapped.

```
[18]: # Evaluate the polar stress components on the 2D mesh (values in MPa) and
# sample them along radial lines at theta = 0 and theta = 90 degrees.
ms_last.point_data["sigma_polar"] = stress_polar.transform(ms_last)

r_probe = np.linspace(TUNNEL_RADIUS, DOMAIN_EXTENT, 35)

def kirsch_polar(r, theta, s_far=-VERTICAL_STRESS / 1.0e6, a=TUNNEL_RADIUS):
    """Kirsch solution for a uniaxial far-field stress s_far in y-direction."""
    cos_2t = np.cos(2.0 * theta)
    sigma_rr = 0.5 * s_far * (1.0 - a**2 / r**2) - 0.5 * s_far * (
        1.0 - 4.0 * a**2 / r**2 + 3.0 * a**4 / r**4
    ) * cos_2t
    sigma_tt = 0.5 * s_far * (1.0 + a**2 / r**2) + 0.5 * s_far * (
```

```

        1.0 + 3.0 * a**4 / r**4
    ) * cos_2t
    return sigma_rr, sigma_tt

fig, ax = plt.subplots(2, 2, figsize=(12, 8), sharex=True)
variables = [
    ot.variables.stress["xx"], # ax[0, 0]
    ot.variables.stress["yy"], # ax[0, 1]
    stress_polar["rr"], # ax[1, 0]
    stress_polar["tt"], # ax[1, 1]
]
line_styles = [("0", 0.0, "C2", "s"), ("90", np.pi / 2, "C3", "D")]
for deg, theta, color, marker in line_styles:
    line_points = np.c_[
        TUNNEL_CENTER_X + r_probe * np.cos(theta),
        TUNNEL_CENTER_Y + r_probe * np.sin(theta),
        np.zeros_like(r_probe),
    ]
    line = pv.PolyData(line_points).sample(ms_last)
    sigma_rr_ana, sigma_tt_ana = kirsch_polar(r_probe, theta)
    numerical = [
        line["sigma"][:, 0] / 1.0e6,
        line["sigma"][:, 1] / 1.0e6,
        line["sigma_polar"][:, 0],
        line["sigma_polar"][:, 1],
    ]
    # On the symmetry lines sigma_rt = 0, so the Cartesian components
    # coincide with the polar ones (swapped on the vertical line).
    if theta == 0.0:
        analytical = [sigma_rr_ana, sigma_tt_ana, sigma_rr_ana, sigma_tt_ana]
    else:
        analytical = [sigma_tt_ana, sigma_rr_ana, sigma_rr_ana, sigma_tt_ana]
    for axis, num, ana in zip(ax.ravel(), numerical, analytical):
        axis.plot(
            r_probe,
            num,
            marker=marker,
            markersize=5,
            linestyle="None",
            color=color,
            label=rf"OGS, $\theta = {deg}^\circ$",
        )
    axis.plot(
        r_probe,
        ana,
        linewidth=0.8,

```

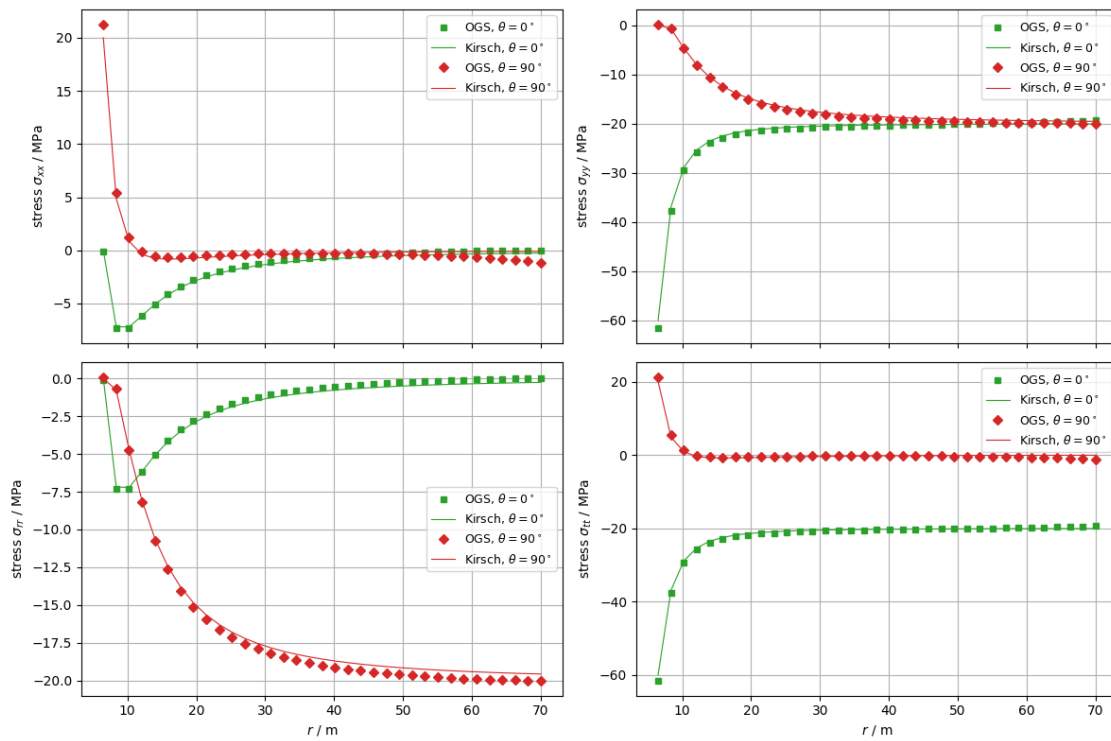
```

        color=color,
        label=rf"Kirsch, $\theta$ = {deg}\textcircled{$}",
    )

for axis, variable in zip(ax.ravel(), variables):
    axis.set_ylabel(variable.get_label())
    axis.grid(True)
    axis.legend(fontsize=9)
for axis in ax[1]:
    axis.set_xlabel("$r$ / m")

fig.tight_layout()
plt.show()

```



5.2 Standard approach

```

[19]: pvd_std = runBenchmark(
    project_file=release_project_file,
    out_dir=out_dir,
    prj_dir=prj_dir,
    output_prefix="kirsch_standard",
    mesh_dir=mesh_dir_quad,
    order=2,
    standard_approach=True,

```

```
)
```

Output prefix is kirsch_standard

OGS executable: /home/mehran/.venvs/ogs312/bin/ogs

```
[20]: ms_std = ot.MeshSeries(pvd_std).scale(time="d")
      extracted_ms_std = ot.MeshSeries.probe(ms_std, probes)
```

100%| | 2/2 [00:00<00:00, 88.84it/s]

```
[21]: pvd_std_linear = runBenchmark(
      project_file=release_project_file,
      out_dir=out_dir,
      prj_dir=prj_dir,
      output_prefix="kirsch_standard_linear",
      mesh_dir=mesh_dir_line,
      order=1,
      standard_approach=True,
      )
```

Output prefix is kirsch_standard_linear

OGS executable: /home/mehran/.venvs/ogs312/bin/ogs

```
[22]: ms_std_linear = ot.MeshSeries(pvd_std_linear).scale(time="d")
      extracted_ms_std_linear = ot.MeshSeries.probe(ms_std_linear, probes)
```

100%| | 2/2 [00:00<00:00, 193.43it/s]

5.3 Automated verification checks The release and standard formulations must produce the same final total stress. Their displacement fields are not compared directly: the release model uses the in-situ state as its zero-displacement reference, whereas the standard model also contains deformation caused by applying the far-field load.

```
[23]: def max_stress_difference_mpa(release_series, standard_series):
      difference = release_series[-1].point_data["sigma"] - standard_series[-1].
      ↪point_data["sigma"]
      return np.max(np.abs(difference)) / 1.0e6

quadratic_stress_difference = max_stress_difference_mpa(ms, ms_std)
linear_stress_difference = max_stress_difference_mpa(ms_linear, ms_std_linear)
initial_displacement = np.max(np.abs(ms[0].point_data["displacement"]))
displacement_reference_difference = np.max(
    np.abs(ms[-1].point_data["displacement"] - ms_std[-1].
    ↪point_data["displacement"]))
)

np.testing.assert_allclose(
    ms[-1].point_data["sigma"],
```

```

    ms_std[-1].point_data["sigma"],
    rtol=1.0e-10,
    atol=1.0e-3,
)
np.testing.assert_allclose(
    ms_linear[-1].point_data["sigma"],
    ms_std_linear[-1].point_data["sigma"],
    rtol=1.0e-10,
    atol=1.0e-3,
)
assert initial_displacement < 1.0e-12

wall_rr_quadratic = extracted_ms[-1].point_data["sigma"][0, 0] / 1.0e6
wall_rr_linear = extracted_ms_linear[-1].point_data["sigma"][0, 0] / 1.0e6
print(f"Maximum final stress difference, quadratic:␣
    ↳{quadratic_stress_difference:.3e} MPa")
print(f"Maximum final stress difference, linear:      {linear_stress_difference:.
    ↳3e} MPa")
print(f"Maximum initial displacement:                {initial_displacement:.3e}␣
    ↳m")
print(f"Expected displacement-reference difference:␣
    ↳{displacement_reference_difference:.4f} m")
print(f"Wall sigma_rr, quadratic:                    {wall_rr_quadratic:.3f}␣
    ↳MPa")
print(f"Wall sigma_rr, linear:                        {wall_rr_linear:.3f} MPa")

```

```

Maximum final stress difference, quadratic: 9.868e-11 MPa
Maximum final stress difference, linear:    4.647e-12 MPa
Maximum initial displacement:              0.000e+00 m
Expected displacement-reference difference: 0.2548 m
Wall sigma_rr, quadratic:                  -0.133 MPa
Wall sigma_rr, linear:                    -2.620 MPa

```

1.1.6 6. Verification and result comparison

```

[24]: fig, ax = plt.subplots(1, 2, figsize=(12, 4))
      series = [
          (extracted_ms[-1], ax[0], "C2", "s", "Release nodal force (quadratic)"),
          (extracted_ms_std[-1], ax[0], "C4", "h", "Standard (quadratic)"),
          (extracted_ms_linear[-1], ax[1], "C3", "D", "Release nodal force (linear)"),
          (extracted_ms_std_linear[-1], ax[1], "C1", "o", "Standard (linear)"),
      ]
      for line_mesh, axis, color, marker, label in series:
          ot.plot.lineplots.line(
              line_mesh,
              ot.variables.stress["xx"],
              ax=axis,

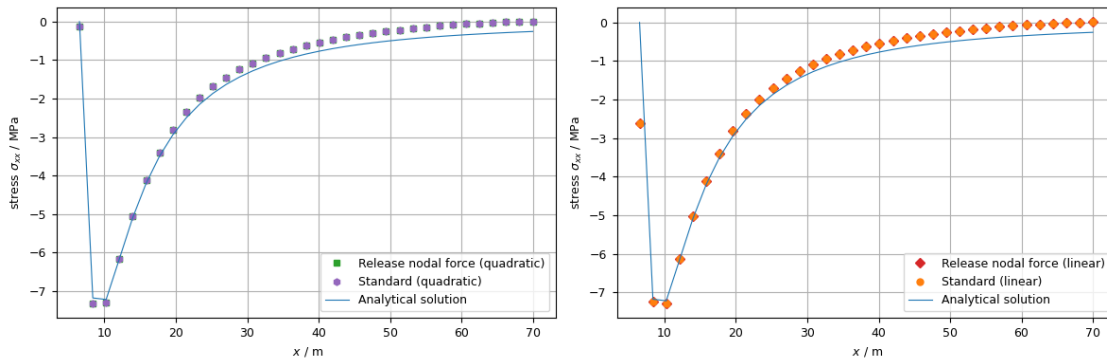
```

```

        fontsize=9,
        grid=False,
        marker=marker,
        markersize=5,
        linestyle="None",
        color=color,
        label=label,
    )

# Kirsch solution: sigma_xx = sigma_rr along theta = 0.
sigma_analytical = kirsch_polar(xs, 0.0)[0]
for axis in ax:
    axis.plot(
        xs,
        sigma_analytical,
        linewidth=0.8,
        color="C0",
        label="Analytical solution",
    )
axis.grid(True)
axis.legend(fontsize=9)
plt.show()

```



The release and standard approaches coincide for both element orders. At the wall the analytical radial stress is zero. The quadratic extrapolated value is about -0.13 MPa, while the linear value is about -2.62 MPa for this mesh. This is a nodal stress-extrapolation and discretization error, not a violation of the imposed traction condition at the integration level.

```

[25]: fig, ax = plt.subplots(1, 2, figsize=(12, 4))
series = [
    (extracted_ms[-1], ax[0], "C2", "s", "Release nodal force (quadratic)"),
    (extracted_ms_std[-1], ax[0], "C4", "h", "Standard (quadratic)"),
    (extracted_ms_linear[-1], ax[1], "C3", "D", "Release nodal force (linear)"),
    (extracted_ms_std_linear[-1], ax[1], "C1", "o", "Standard (linear)"),

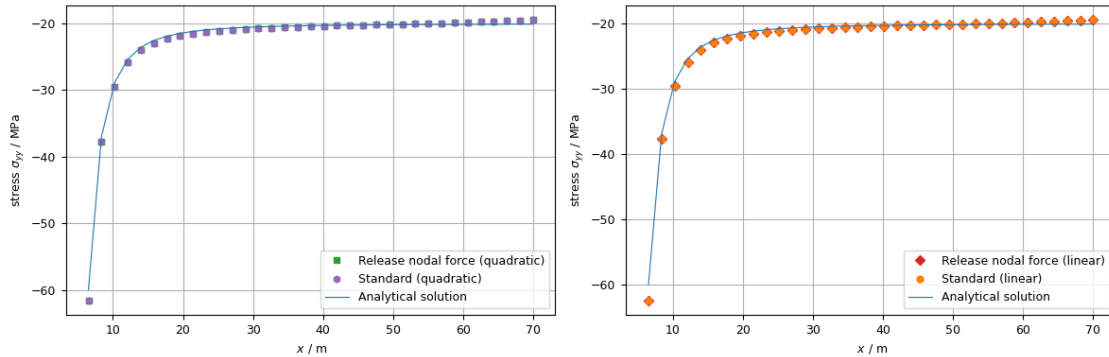
```

```

]
for line_mesh, axis, color, marker, label in series:
    ot.plot.lineplots.line(
        line_mesh,
        ot.variables.stress["yy"],
        ax=axis,
        fontsize=9,
        grid=False,
        marker=marker,
        markersize=5,
        linestyle="None",
        color=color,
        label=label,
    )

# Kirsch solution:  $\sigma_{yy} = \sigma_{tt}$  along  $\theta = 0$ .
sigma_analytical = kirsch_polar(xs, 0.0)[1]
for axis in ax:
    axis.plot(
        xs,
        sigma_analytical,
        linewidth=0.8,
        color="C0",
        label="Analytical solution",
    )
    axis.grid(True)
    axis.legend(fontsize=9)
plt.show()

```



The tangential stress concentration is captured well by both formulations. Quadratic elements give the more accurate extrapolated wall value. Differences near the outer boundary are caused mainly by replacing the infinite domain with a finite square boundary.

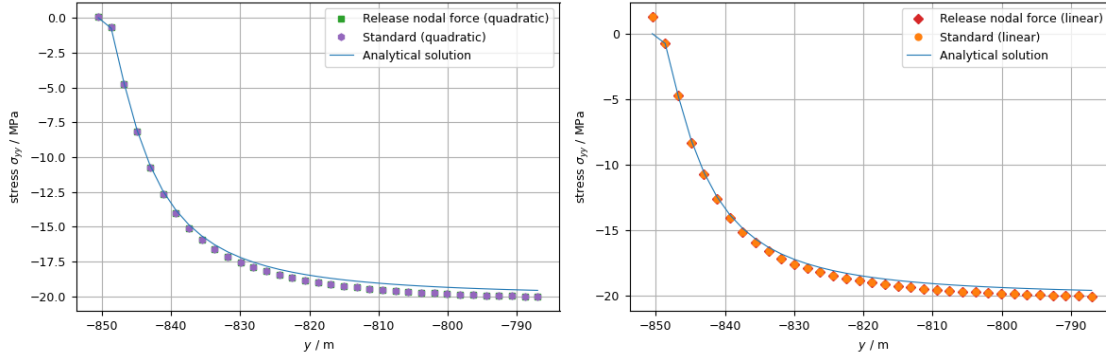
```

[26]: r_vertical = np.linspace(TUNNEL_RADIUS, DOMAIN_EXTENT, 35)
ys = TUNNEL_CENTER_Y + r_vertical
y_axis = np.c_[
    np.full_like(ys, TUNNEL_CENTER_X),
    ys,
    np.zeros_like(ys),
]
extracted_ms_y = ot.MeshSeries.probe(ms, y_axis)
extracted_ms_y_linear = ot.MeshSeries.probe(ms_linear, y_axis)
extracted_ms_std_y = ot.MeshSeries.probe(ms_std, y_axis)
extracted_ms_std_y_linear = ot.MeshSeries.probe(ms_std_linear, y_axis)

[27]: fig, ax = plt.subplots(1, 2, figsize=(12, 4))
series = [
    (extracted_ms_y[-1], ax[0], "C2", "s", "Release nodal force (quadratic)"),
    (extracted_ms_std_y[-1], ax[0], "C4", "h", "Standard (quadratic)"),
    (extracted_ms_y_linear[-1], ax[1], "C3", "D", "Release nodal force_␣
↪(linear)"),
    (extracted_ms_std_y_linear[-1], ax[1], "C1", "o", "Standard (linear)"),
]
for line_mesh, axis, color, marker, label in series:
    ot.plot.lineplots.line(
        line_mesh,
        ot.variables.stress["yy"],
        ax=axis,
        fontsize=9,
        grid=False,
        marker=marker,
        markersize=5,
        linestyle="None",
        color=color,
        label=label,
    )

# Kirsch solution: sigma_yy = sigma_rr along theta = 90.
sigma_analytical = kirsch_polar(r_vertical, np.pi / 2)[0]
for axis in ax:
    axis.plot(
        ys,
        sigma_analytical,
        linewidth=0.8,
        color="C0",
        label="Analytical solution",
    )
    axis.grid(True)
    axis.legend(fontsize=9)
plt.show()

```



In the two figures above, the radial stress σ_r profiles along the $\theta = 90^\circ$ axis are similarly compared across the standard approach, the release nodal force approach, and the analytical solution. The same conclusion as above can be drawn.

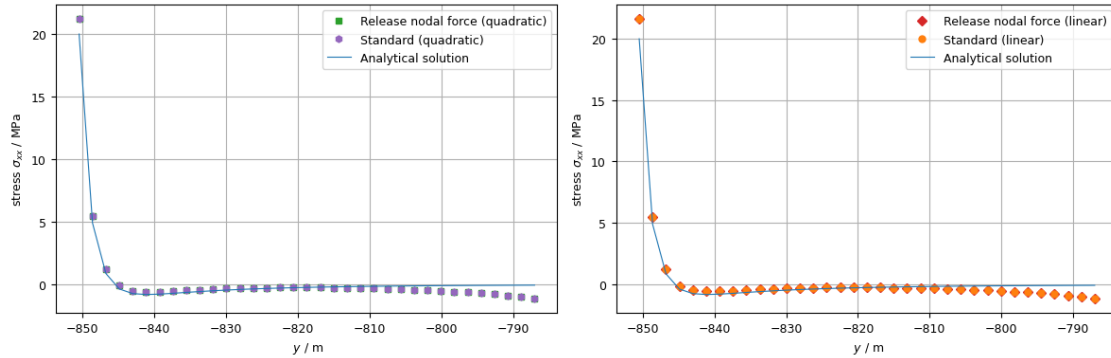
```
[28]: fig, ax = plt.subplots(1, 2, figsize=(12, 4))
series = [
    (extracted_ms_y[-1], ax[0], "C2", "s", "Release nodal force (quadratic)"),
    (extracted_ms_std_y[-1], ax[0], "C4", "h", "Standard (quadratic)"),
    (extracted_ms_y_linear[-1], ax[1], "C3", "D", "Release nodal force_
↪(linear)"),
    (extracted_ms_std_y_linear[-1], ax[1], "C1", "o", "Standard (linear)"),
]
for line_mesh, axis, color, marker, label in series:
    ot.plot.lineplots.line(
        line_mesh,
        ot.variables.stress["xx"],
        ax=axis,
        fontsize=9,
        grid=False,
        marker=marker,
        markersize=5,
        linestyle="None",
        color=color,
        label=label,
    )

# Kirsch solution: sigma_xx = sigma_tt along theta = 90.
sigma_analytical = kirsch_polar(r_vertical, np.pi / 2)[1]
for axis in ax:
    axis.plot(
        ys,
        sigma_analytical,
        linewidth=0.8,
        color="C0",
```

```

        label="Analytical solution",
    )
    axis.grid(True)
    axis.legend(fontsize=9)
plt.show()

```



In the two figures above, the tangential stress σ_θ profiles along the $\theta = 90^\circ$ axis—obtained using the standard approach, the release nodal force approach, and the analytical solution—are compared. The figure on the right shows results using linear elements, while the figure on the left displays results using quadratic elements.

The figures show that the quadratic elements deliver more accurate tangential stress results than the linear elements do.

```

[29]: history_radii = TUNNEL_RADIUS + np.array([0.0, 1.0, 2.5, 3.5, 8.5])
      spoints = np.c_[
          np.full_like(history_radii, TUNNEL_CENTER_X),
          TUNNEL_CENTER_Y + history_radii,
          np.zeros_like(history_radii),
      ]
      labels = [f"r = {radius:.1f} m" for radius in history_radii]

```

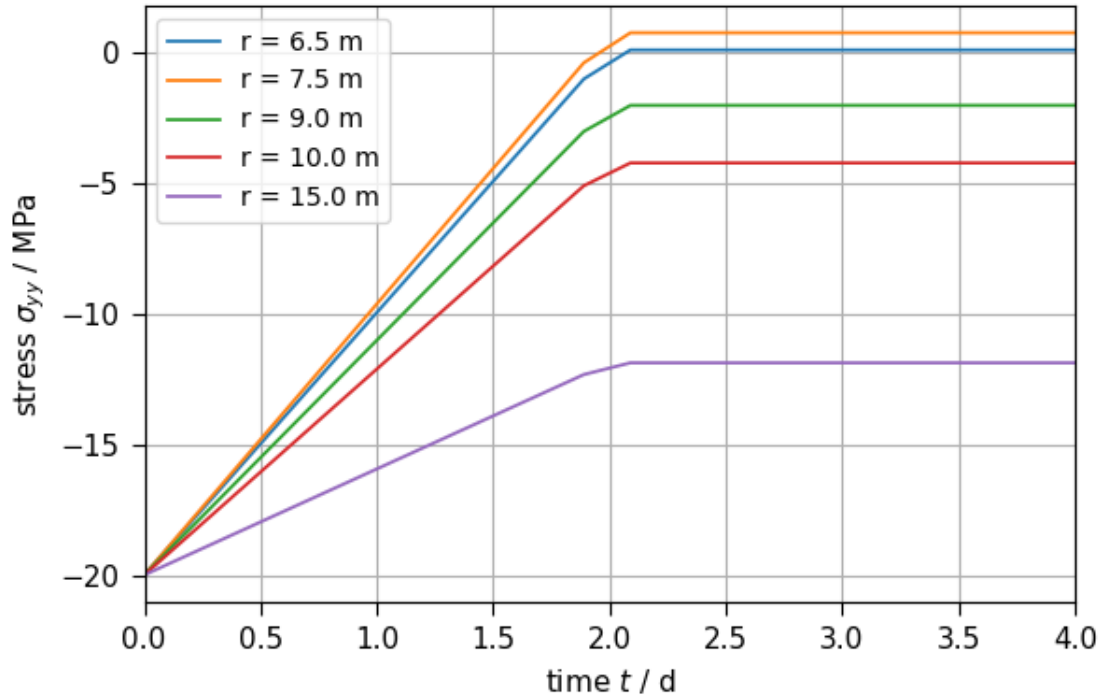
```

[30]: extracted_sp = ot.MeshSeries.probe(ms, spoints)
      ot.plot.line(
          extracted_sp,
          "time",
          ot.variables.stress["yy"],
          labels=labels,
          figsize=(6, 4),
          fontsize=11,
          grid=False,
          linewidth=0.5,
      )
      plt.xlim(0, 4)

```

```
plt.legend()

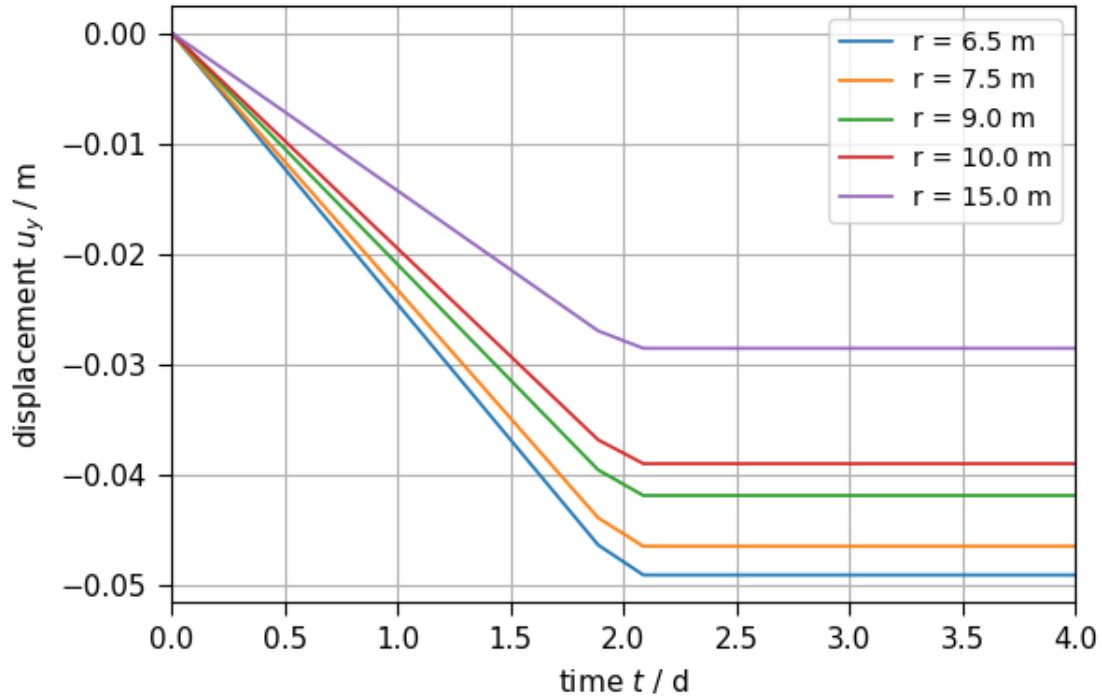
plt.grid(True)
plt.show()
```



The figure shows radial stress along the vertical symmetry line ($\theta = 90^\circ$). At the wall it evolves from the initial compressive stress toward the traction-free value as the support traction is released. The response becomes constant after the two-day excavation interval.

```
[31]: ot.plot.line(
    extracted_sp,
    "time",
    ot.variables.displacement["y"],
    labels=labels,
    figsize=(6, 4),
    fontsize=11,
    grid=False,
    linewidth=0.5,
)
plt.xlim(0, 4)
plt.legend()

plt.grid(True)
plt.show()
```



Along $\theta = 90^\circ$, the y displacement is radial. Its magnitude grows during unloading and then remains constant. These displacements are measured relative to the initially stressed in-situ state, which is why they should not be compared directly with the standard model's total displacement.

1.1.7 7. Element-deactivation representation

The alternative mesh contains a quarter-disc inside the opening. Those elements (material ID 1) are inactive for the entire simulation; this is a representation check, not a staged active-to-inactive excavation. OGS writes zero stress for the inactive cells. At nodes shared by active and inactive cells, nodal stress extrapolation therefore mixes physical rock stresses with zeros. Exact wall-node stresses from this mesh must not be used for verification. The comparison below excludes that shared node and evaluates the active rock away from the interface.

```
[32]: if not gmsh.isInitialized():
    gmsh.initialize(readConfigFiles=False)

gmsh.model.add("Mesh_with_inactive_cavern")
mesh_generator = MeshGenerator(
    gmsh_model=gmsh.model,
    tunnel_radius_i=TUNNEL_RADIUS,
    tunnel_radius_o=MESH_TRANSITION_WIDTH,
    tunnel_center_x=TUNNEL_CENTER_X,
    tunnel_center_y=TUNNEL_CENTER_Y,
    width=DOMAIN_EXTENT,
```

```

        height=DOMAIN_EXTENT,
        width_m=INNER_BLOCK_EXTENT,
        height_m=INNER_BLOCK_EXTENT,
    )
    mesh_dir_with_hole = Path(out_dir, "Mesh", "Entire")
    mesh_generator.generate_meshes(out_dir=mesh_dir_with_hole, with_cavern=True,
    ↪order=2)

    gmsh.finalize()

```

```

INFO:ogstools.meshes.gmsh_converter:Detected domain dimension of 2
INFO:ogstools.meshes.gmsh_converter:Found material IDs: [1 2]
INFO:ogstools.meshes.gmsh_converter:Renumbered to: [0 1]
INFO:ogstools.meshes.gmsh_converter:domain: UnstructuredGrid (0x79b8f4bfbfa0)
  N Cells:      3148
  N Points:     3082
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:    -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      1
INFO:ogstools.meshes.gmsh_converter:bottom: UnstructuredGrid (0x79b8f49c58a0)
  N Cells:      67
  N Points:     68
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:    -8.570e+02, -8.570e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      2
INFO:ogstools.meshes.gmsh_converter:left: UnstructuredGrid (0x79b8f49c79a0)
  N Cells:      67
  N Points:     68
  X Bounds:     0.000e+00, 0.000e+00
  Y Bounds:    -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      2
INFO:ogstools.meshes.gmsh_converter:right: UnstructuredGrid (0x79b8f4c5d9c0)
  N Cells:      40
  N Points:     41
  X Bounds:     7.000e+01, 7.000e+01
  Y Bounds:    -8.570e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      2
INFO:ogstools.meshes.gmsh_converter:top: UnstructuredGrid (0x79b8f80b2740)
  N Cells:      40
  N Points:     41
  X Bounds:     0.000e+00, 7.000e+01
  Y Bounds:    -7.870e+02, -7.870e+02
  Z Bounds:     0.000e+00, 0.000e+00
  N Arrays:      2

```

```

INFO:ogstools.meshes.gmsh_converter:arc: UnstructuredGrid (0x79b8f44c3220)
  N Cells:    40
  N Points:   41
  X Bounds:   0.000e+00, 6.500e+00
  Y Bounds:   -8.570e+02, -8.505e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   3
INFO:ogstools.meshes.gmsh_converter:cavern: UnstructuredGrid (0x79b8f44c1060)
  N Cells:    348
  N Points:   202
  X Bounds:   0.000e+00, 6.500e+00
  Y Bounds:   -8.570e+02, -8.505e+02
  Z Bounds:   0.000e+00, 0.000e+00
  N Arrays:   2
INFO:ogstools.meshes.gmsh_converter:Conversion complete.

```

```

Info    : Meshing 1D...
Info    : [ 0%] Meshing curve 1 (Line)
Info    : [ 10%] Meshing curve 2 (Line)
Info    : [ 10%] Meshing curve 3 (Line)
Info    : [ 20%] Meshing curve 4 (Line)
Info    : [ 20%] Meshing curve 5 (Line)
Info    : [ 30%] Meshing curve 6 (Line)
Info    : [ 30%] Meshing curve 7 (Line)
Info    : [ 40%] Meshing curve 8 (Line)
Info    : [ 40%] Meshing curve 9 (Line)
Info    : [ 50%] Meshing curve 10 (Line)
Info    : [ 50%] Meshing curve 11 (Line)
Info    : [ 60%] Meshing curve 12 (Line)
Info    : [ 60%] Meshing curve 13 (Line)
Info    : [ 60%] Meshing curve 14 (Line)
Info    : [ 70%] Meshing curve 15 (Line)
Info    : [ 70%] Meshing curve 16 (Line)
Info    : [ 80%] Meshing curve 17 (Circle)
Info    : [ 80%] Meshing curve 18 (Circle)
Info    : [ 90%] Meshing curve 19 (Circle)
Info    : [ 90%] Meshing curve 20 (Circle)
Info    : [100%] Meshing curve 21 (Line)
Info    : [100%] Meshing curve 22 (Line)
Info    : Done meshing 1D (Wall 0.000552822s, CPU 0.000838s)
Info    : Meshing 2D...
Info    : [ 0%] Meshing surface 1 (Transfinite)
Info    : [ 20%] Meshing surface 2 (Transfinite)
Info    : [ 30%] Meshing surface 3 (Transfinite)
Info    : [ 40%] Meshing surface 4 (Transfinite)
Info    : [ 60%] Meshing surface 5 (Transfinite)
Info    : [ 70%] Meshing surface 6 (Transfinite)
Info    : [ 80%] Meshing surface 7 (Transfinite)

```

```

Info      : [ 90%] Meshing surface 8 (Plane, Delaunay)
Info      : Done meshing 2D (Wall 0.00243607s, CPU 0.001907s)
Info      : 3082 nodes 3577 elements
Info      : Writing '_out/Mesh/Entire/domain.msh'...
Info      : Done writing '_out/Mesh/Entire/domain.msh'

info: Reordering nodes...
info: Method: Reversing order of nodes unless it is considered correct by the
OGS6 standard, i.e. such that  $\det(J) > 0$ , where J is the Jacobian of the global-
to-local coordinate transformation.
info: Corrected 348 elements.
info: VTU file written.
domain: 3148 cells
bottom: 67 cells
left: 67 cells
right: 40 cells
top: 40 cells
arc: 40 cells
cavern: 348 cells
info: Reordering nodes...
info: Method: Reversing order of nodes unless it is considered correct by the
OGS6 standard, i.e. such that  $\det(J) > 0$ , where J is the Jacobian of the global-
to-local coordinate transformation.
info: Corrected 0 elements.
info: VTU file written.
Create quadratic mesh for _out/Mesh/Entire/left.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/cavern.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/top.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/arc.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/bottom.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/right.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
Create quadratic mesh for _out/Mesh/Entire/domain.vtu
info: Create a quadratic order mesh
info: Save the new mesh into a file
info: Mesh reading time: 0.00988916 s
info: MeshNodeSearcher construction time: 0.000343399 s

```

```

info: identifySubdomainMesh(): identifySubdomainMeshNodes took 0.000654475 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00290755 s
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.3919e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00573443 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'bottom' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 3.0549e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00565512 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'left' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.224e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00605259 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'right' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 1.3098e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00586588 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'top' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 2.3736e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.0058779 s
info: There is already a 'number_bulk_elements' property present in the
subdomain mesh 'arc' and it is equal to the newly computed values.
info: identifySubdomainMesh(): identifySubdomainMeshNodes took 7.8419e-05 s
info: identifySubdomainMesh(): identifySubdomainMeshElements took 0.00578647 s
info: There is already a 'bulk_element_ids' property present in the subdomain
mesh 'cavern' and it is equal to the newly computed values.
info: identifySubdomains time: 0.0392652 s
info: writing time: 0.00661543 s
info: Entire run time: 0.0561843 s

```

```

[33]: pvd_ed = runBenchmark(
    project_file=release_project_file,
    out_dir=out_dir,
    prj_dir=prj_dir,
    output_prefix="kirsch_element_deactivation",
    mesh_dir=mesh_dir_with_hole,
    order=2,
    use_element_deactivation=True,
)

```

Output prefix is kirsch_element_deactivation
OGS executable: /home/mehran/.venvs/ogs312/bin/ogs

```

[34]: ms_ed = ot.MeshSeries(pvd_ed).scale(time="d")
    extracted_ms_ed_x = ot.MeshSeries.probe(ms_ed, probes)
    extracted_ms_ed_y = ot.MeshSeries.probe(ms_ed, y_axis)

```

100% | 22/22 [00:00<00:00, 44.00it/s]

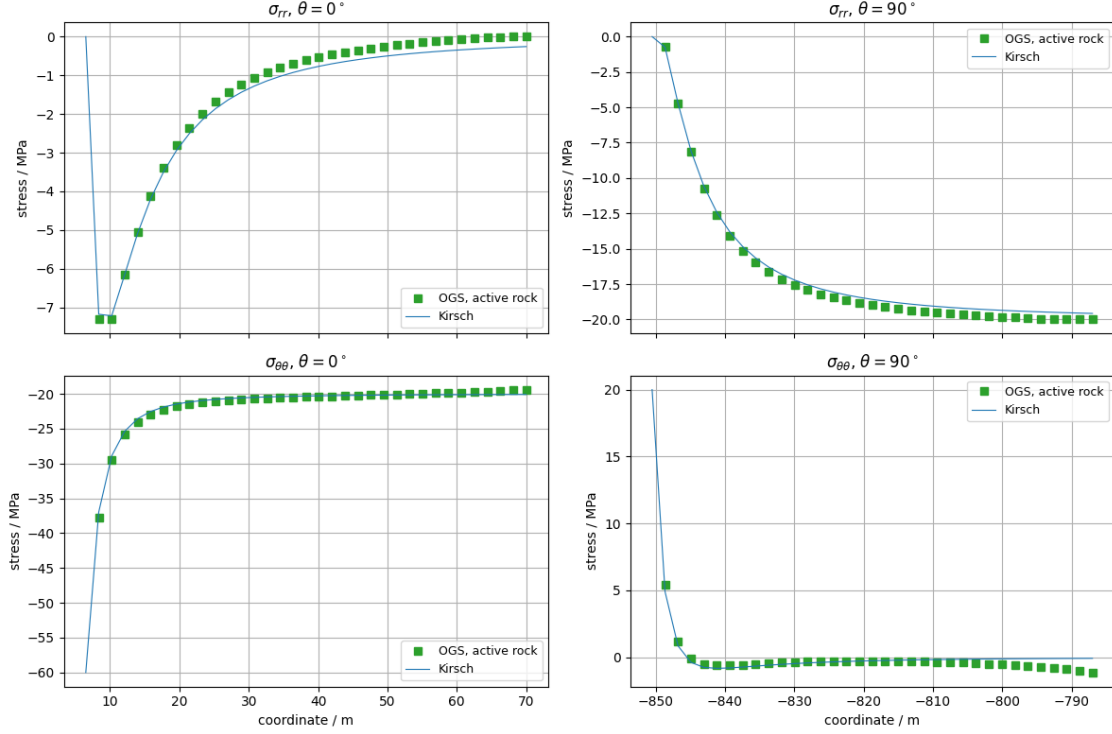
```

[35]: # Skip index 0: it is the node shared by active and inactive elements.
rock_slice = slice(1, None)
ed_x = extracted_ms_ed_x[-1].point_data["sigma"] / 1.0e6
ed_y = extracted_ms_ed_y[-1].point_data["sigma"] / 1.0e6
release_x = extracted_ms[-1].point_data["sigma"] / 1.0e6
release_y = extracted_ms_y[-1].point_data["sigma"] / 1.0e6

np.testing.assert_allclose(ed_x[rock_slice, :2], release_x[rock_slice, :2],
    ↪atol=1.0e-8)
np.testing.assert_allclose(ed_y[rock_slice, :2], release_y[rock_slice, :2],
    ↪atol=1.0e-8)

fig, ax = plt.subplots(2, 2, figsize=(12, 8), sharex="col")
profiles = [
    (xs, ed_x[:, 0], kirsch_polar(xs, 0.0)[0], r"$\sigma_{rr}$",
    ↪r"$\theta=0^\circ$"),
    (ys, ed_y[:, 1], kirsch_polar(r_vertical, np.pi / 2)[0], r"$\sigma_{rr}$",
    ↪r"$\theta=90^\circ$"),
    (xs, ed_x[:, 1], kirsch_polar(xs, 0.0)[1], r"$\sigma_{\theta\theta}$",
    ↪r"$\theta=0^\circ$"),
    (ys, ed_y[:, 0], kirsch_polar(r_vertical, np.pi / 2)[1],
    ↪r"$\sigma_{\theta\theta}$", r"$\theta=90^\circ$"),
]
for axis, (coordinates, numerical, analytical, title) in zip(ax.ravel(),
    ↪profiles):
    axis.plot(coordinates[rock_slice], numerical[rock_slice], "s", color="C2",
    ↪label="OGS, active rock")
    axis.plot(coordinates, analytical, color="C0", linewidth=0.8,
    ↪label="Kirsch")
    axis.set_title(title)
    axis.set_ylabel("stress / MPa")
    axis.grid(True)
    axis.legend(fontsize=9)
for axis in ax[1]:
    axis.set_xlabel("coordinate / m")
fig.tight_layout()
plt.show()

```



Away from the node shared by the active and deactivated elements, the element-deactivation model and the hole-only model produce identical in-plane stresses for this mesh. The stress value at the shared tunnel-wall node is excluded because nodal stress extrapolation combines contributions from both active and deactivated elements, making it a postprocessing artifact. Consequently, a radial stress close to zero at this node is not, by itself, sufficient to validate the element-deactivation result. Validation should instead compare stresses at nodes or integration points influenced only by active elements.

1.1.8 8. Conclusions

- The release-nodal-force and standard formulations reproduce the same final total stress tensor when the plane-strain initial stress includes $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$.
- Quadratic elements improve the extrapolated wall stresses on this mesh. This does not imply that linear elements are generally unsuitable; mesh refinement and evaluation at integration points must also be considered.
- Small far-field differences are expected because the numerical model is finite whereas the Kirsch solution assumes an infinite domain.
- Element deactivation is compatible with the release approach, but extrapolated stresses at nodes shared with inactive cells are not suitable verification quantities.
- Release-model displacement is excavation-induced relative to the initialized in-situ state; standard-model displacement uses a different reference state.

1.2 References

[1] E. G. Kirsch (1898), “Die Theorie der Elastizität und die Bedürfnisse der Festigkeitslehre,” *Zeitschrift des Vereines Deutscher Ingenieure*, vol. 42, pp. 797–807.

[2] OpenGeoSys documentation: <https://www.opengeosys.org/docs/>